# Prayas JEE AIR (2025)

## WEEKLY TEST 29

**DURATION : 180 Minutes** | **DATE : 11/01/2025** | **M. MARKS : 300**

# ANSWER KEY

**PHYSICS**

1. (3)
2. (1)
3. (1)
4. (1)
5. (3)
6. (3)
7. (1)
8. (4)
9. (4)
10. (2)
11. (1)
12. (4)
13. (4)
14. (2)
15. (2)
16. (2)
17. (3)
18. (1)
19. (3)
20. (2)
21. (16)
22. (2)
23. (25)
24. (9)
25. (2)

**CHEMISTRY**

26. (3)
27. (4)
28. (4)
29. (2)
30. (4)
31. (2)
32. (2)
33. (2)
34. (1)
35. (2)
36. (4)
37. (3)
38. (2)
39. (4)
40. (3)
41. (3)
42. (1)
43. (3)
44. (4)
45. (3)
46. (25)
47. (132)
48. (0)
49. (5)
50. (241)

**MATHEMATICS**

51. (1)
52. (1)
53. (2)
54. (3)
55. (3)
56. (3)
57. (4)
58. (2)
59. (4)
60. (1)
61. (4)
62. (1)
63. (2)
64. (1)
65. (3)
66. (3)
67. (3)
68. (1)
69. (4)
70. (3)
71. (1)
72. (2)
73. (6)
74. (39)
75. (64)

## SECTION-I (PHYSICS)

**1.** **(3)**

Given, $t = mx^2 + nx$

$\frac{dt}{dx} = 2mx + n \Rightarrow \frac{dx}{dt} = \frac{1}{2mx+n}$

$v = \frac{1}{2mx+n}$

Retardation of the motion

$a = v\frac{dv}{dx} = v\frac{-2m}{(2mx+n)^2} = -2mv^3$

$m = 3$

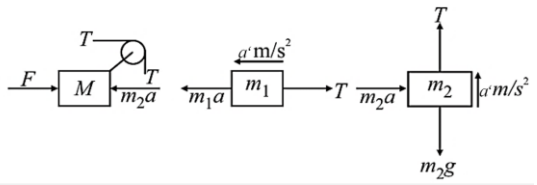
$n = 2$

and at $x = \frac{1}{3}$.

$v = \frac{1}{4}$

$a = \frac{-3}{64}$ m/sec$^2$

**2.** **(1)**




$F - T = (m_2 + M)a \quad ..(1)$

$m_1a - T = m_1a'$

$T - m_2g = m_2a'$

$a' = \frac{m_1F - m_2g(m_1 + m_2 + M)}{((m_1 + m_2)(M + m_2) + m_1m_2)}$

$a' = 2\,\text{m/sec}^2$

**3.** **(1)**

Power, $P =$ constant

$P = Fv = mv\frac{dv}{dx}v$

$\frac{P}{m}\int dx = \int v^2 dv$

$\frac{Px}{m} = \frac{v^3}{3}$

$\left(\frac{3P}{m}x\right)^{1/3} = v = \frac{dx}{dt}$

$\left(\frac{3P}{m}\right)^{1/3}\int_0^t dt = \int_0^x x^{-1/3}dx$

$\Rightarrow x = \left(\frac{8P}{9m}\right)^{1/2} t^{3/2}$

$P$ = 60 W

$m$ = 90 kg.

$t$ = 3 sec.

$x$ = 4 m.

**4.** **(1)**

Total distance,

$d = h + 2e^2h + 2e^4h + 2e^6h + 2e^8h + \ldots..$

$d = h + 2e^2h(1 + e^2 + e^4 + e^6 + \ldots..)$

$d = \frac{(1-e^2)h + 2e^2h}{(1-e^2)} = \frac{h(1+e^2)}{1-e^2}$

Total time $= T + 2eT + 2e^2T + 2e^3T + \ldots..$

Total time $= T + 2eT(1 + e + e^2 + e^3 + \ldots..)$

$= T + 2e.T\left(\frac{1}{1-e}\right)$

Total time $= \frac{T(1+e)}{1-e}$

Average speed of the ball $= \frac{h}{T}\left(\frac{1+e^2}{(1+e)(1-e)}\frac{(1-e)}{(1+e)}\right)$

$V_{avg} = \frac{\sqrt{\frac{gh}{2}}(1+e^2)}{(1+e)^2}$

$\because \quad h^1 = e^2h$

$\frac{64}{100} = e^2 \Rightarrow e = \frac{8}{10} = 0.8$

$V_{avg} = \frac{9(1+(0.8)^2)}{\sqrt{2}(1+0.8)^2} = 3.2\,\text{m/sec.}$

**5.** **(3)**

Mass of material $M = (\pi R^2 L\rho)$

$\therefore \quad I = M\left(\frac{R^2}{4} + \frac{L^2}{12}\right) = M\left(\frac{M}{4\pi\rho L} + \frac{L^2}{12}\right)$

$\therefore$ For I to be minimum

$= \frac{dI}{dL} = 0 = M\left(-\frac{M}{4\pi\rho L^2} + \frac{L}{6}\right)$

$\Rightarrow \quad \frac{M}{\pi\rho L} = \frac{4}{6}L^2$

$\Rightarrow \quad R^2 = \frac{2}{3}L^2$

$I_{min} = M\left(\frac{2L^2}{12} + \frac{L^2}{12}\right)$

$I_{min} = \frac{ML^2}{4}$

$Mk^2_{min} = I_{min}$

$k_{min} = L/2$

**6.** **(3)**

At = $d = R/2$

$$g_1 = \frac{GMr}{R^3} = \left(\frac{gR^2}{R^3}\right)(R/2) = g/2$$

$$g(h) = \frac{G\,\mathrm{m}}{r^2} = g/2$$

$$r = \sqrt{2}R$$

$$h = (\sqrt{2}-1)R$$

$$\frac{h}{R} = 0.41$$

**7.** **(1)**

So force on each columns will be

$$= \frac{mg}{4} = 125\times10^3\ \mathrm{N}$$

Thus Strain $\varepsilon = \dfrac{F}{AY}$

$$\delta_\varepsilon = \frac{\sigma^2}{2Y}$$

$$\delta_\varepsilon = \frac{m^2g^2}{32A^2Y}$$

$$\delta_\varepsilon = 6.94\,\mathrm{mJ}/m^3$$

**8.** **(4)**

Using equation of continuity, $A_1V_1 = A_2V_2$

$60.4V_A = 30.2V_B \Rightarrow 2V_A = V_B$

Using Bernoullies equation,

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2$$

$$\Rightarrow P_A - P_B = \frac{1}{2}\rho\left(V_B^2 - V_A^2\right)$$

$$\Rightarrow \Delta P = \frac{1}{2}\times1000\left(V_B^2 - \frac{V_B^2}{4}\right)$$

$$\Rightarrow \Delta P = 500\times\frac{3V_B^2}{4}$$

$$\Rightarrow V_B = \sqrt{\frac{(\Delta P)\times4}{1500}} = \sqrt{\frac{375\times4}{1500}}\ \mathrm{m/s} = 1\mathrm{m/sec.}$$

$$= 10^2\ \mathrm{cm/s}$$

Volume flow rate

$$= 30.2\times10^2 \approx 3020\,\mathrm{cm}^3/\mathrm{sec.}$$

**9.** **(4)**

$\Rightarrow Tv^{x-1}$ = constant

$\Rightarrow \quad x = -1$

$$\Rightarrow \Delta W = \frac{nR\Delta T}{1-x} = \frac{RT_0}{2}$$

$$\Rightarrow \Delta U = \frac{3}{2}R\,\mathrm{T}_0$$

$$C = C_v + \frac{R}{1-x}$$

$C = 2R$

$\Rightarrow \Delta Q = 2RT_0$

**10.** **(2)**

**Case 1:** Before collision,

Speed, $v = \sqrt{2gL\left(1-\cos\theta_0\right)}$

**Case 2:** After collision,

Speed, $v_1 = \sqrt{2gL\left(1-\cos\theta_1\right)}$

$$\because \frac{v_1}{v} = \frac{M-m}{M+m}$$

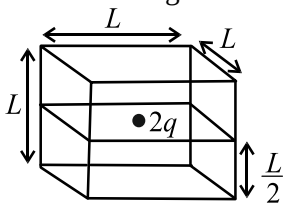
$$\therefore \sqrt{\frac{1-\cos\theta_1}{1-\cos\theta_0}} = \frac{M-m}{M+m}$$

$$\frac{\sin\left(\theta_1/2\right)}{\sin\left(\theta_0/2\right)} = \frac{M-m}{M+m} \quad \left[\because 1-\cos2\theta = 2\sin^2\theta\right]$$

$$\frac{\theta_1}{\theta_0} = \frac{M-m}{M+m} = \frac{1}{3}. \qquad \text{[As angle is small]}$$

**11.** **(1)**

Total electric flux through the cube $\phi = 2q/\varepsilon_0$




Flux passing through the face opposite to surface

$$\phi_1 = \frac{2q}{6\varepsilon_0}$$

$$\phi_2 = \frac{2q}{12\varepsilon_0}$$

$$\phi_1 + \phi_2 = \frac{q}{2\varepsilon_0}$$

**12.** **(4)**

$$\Delta H = \Delta U = \frac{1}{2}\times\frac{C\times3C}{C+3C}\left(V^2\right)$$

$$= \frac{1}{2}\times\frac{3}{4}CV^2 = \frac{3}{8}CV^2$$

$$U_i = \frac{1}{2}CV^2$$

$$f = \frac{H}{U_i} = \frac{3CV^2/8}{CV^2/2} = \frac{3}{4}$$

**13.** **(4)**

$$r_{eq} = \frac{8}{5}\Omega$$

$$i = \frac{8}{(8/5)} = 5\,\mathrm{A}$$

$$P = Vi = 8\times5 = 40\,\mathrm{W}$$

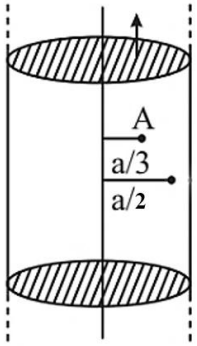
**14.** **(2)**

Magnetic precures

$$P(r)=\frac{\mu_0 J^2}{4}\left(R^2-r^2\right)$$

$$\frac{P(a/3)}{P(a/2)}=\frac{8a^2/9}{3a^2/4}$$

$$=\frac{32}{27}$$

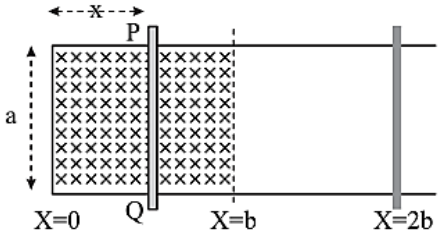



**15.** **(2)**

$$\Delta q=\frac{\Delta\phi}{R}$$

$$\Delta\phi=\int_0^b B(x)bdx$$

$$\Delta\phi=\frac{2B_0b^2}{3}$$




**16.** **(2)**

$$\Rightarrow \sin^{-}\left(\frac{1}{m}\right)<\tan^{-}\left(\frac{\mu}{m}\right)$$

$$\Rightarrow \frac{1}{\sqrt{m^2-1}}<\frac{\mu}{m}$$

$$\mu>\frac{m}{\sqrt{m^2-1}}$$

$$\mu>\frac{4}{\sqrt{7}}$$

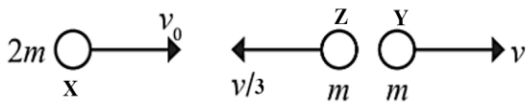
**17.** **(3)**

As indices of refraction changes, wavelength change. And as medium changes, optical path changes.

$\because$ Path difference Phase difference $=\frac{\lambda}{2\pi}\times$

So phase difference changes.

**18.** **(1)**




let mass of *B* and *C* be *m* each. By conservation o momentum. A

$$2mv_0=mv-\frac{mv}{3}$$

$$v=3v_0$$

$$p_A=2mv_0,\ p_B=3mv_0,\ p_c=mv_0$$

De – Broglie wavelength $\lambda=\frac{h}{p}$

$$\lambda_0=\frac{h}{2mv_0};\lambda_y=\frac{h}{3mv_0};\lambda_C=\frac{h}{mv_0}$$

$$\lambda_y+\lambda_z=4h/3mV_0=\frac{8\lambda_0}{3}$$

**19.** **(3)**

$$\frac{-13.6(2)^2}{n^2}=\frac{-13.6(9)}{36}$$

$$\Rightarrow n^2=16$$

$$\Rightarrow n=4$$

**20.** **(2)**

$$B.E.=(\Delta m)C^2$$

$$\Delta m=\left(26m_p+30m_n\right)-\left(m_{Fe}\right)$$

$$=(26\times1.00783+30\times1.00867)-(55.845)$$

B.E. $=\Delta m\times931$MeV

$\Rightarrow$ B.E. $=576$ MeV

$$\Rightarrow \frac{\text{B.E.}}{\text{Nucl.}}=\frac{576\text{ MeV}}{56}=10.3\text{MeV}$$

**21.** **(16)**

$$\theta=\omega_0t+\frac{1}{2}\alpha t^2\Rightarrow\theta=100\text{rad}$$

$\therefore$ Number of revolutions $=\frac{100}{2\pi}=16$ (approx.)

**22.** **(2)**

The resistance of the loops,

$R_1=2\pi r_1\times0.10=2\pi\times0.1\times0.1$

and $R_2=2\pi r_2\times0.1$

Flux in the smaller loop, $\phi=B_2A_1$

$$= \frac{\mu_0 i_2}{2r_2}\pi r_1^2$$

$$= \frac{\mu_0\left[\frac{V}{R_2}\right]\pi r_1^2}{2r_2}$$

$$= \frac{\mu_0 \frac{[4(1+t)]}{R_2}\pi r_1^2}{2r_2}$$

The induced current, $i_1 = \frac{e}{R_I} = \frac{[d\phi/dt]}{R_I}$

After substituting the value and simplifying we get $i = 2\mu A$.

**23. (25)**

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{2}{2\pi\sqrt{5\times 80\times 10^{-6}}} = \frac{25}{\pi}\text{Hz}$$

**24. (9)**

If $v_{max}$ is the speed of the fastest electron emitted from the metal surface, then $\frac{hc}{\lambda} = W_0 + \frac{1}{2}mv_{max^2}$

$$\frac{\left(6.63\times 10^{-34}\right)\times\left(3\times 10^{8}\right)}{\left(155\times 10^{-9}\right)}$$

$$= 2\times\left(1.6\times 10^{-19}\right) + \frac{1}{2}\left(9\times 10^{-31}\right)v^2{}_{max}$$

The radius of the electron is given by

$$r = \frac{mv}{qB} = 9\,\text{mm}$$

**25. (2)**

$\Delta_{temp} = \Delta_{boad}$

and $A = \pi r^2 = \pi\left(\frac{10^{-3}}{\sqrt{\pi}}\right)^2 = 10^{-6}$

$$L\alpha\Delta T = \frac{FL}{AY}$$

$$= \frac{F\times 0.2}{\left(10^{-6}\right)\times 10^{11}} = 0.2\times 10^{-5}\times 20$$

$$\therefore \quad F = 20N \therefore m = \frac{f}{g} = 2 = 2kg$$

## SECTION-II (CHEMISTRY)

**26. (3)**

Test of terminal alkynes.

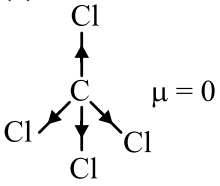
**27. (4)**

Across a period metallic strength decreases & down the group it increases.

**28. (4)**

$$-\frac{d[A_3]}{dt} = \frac{1}{3}\frac{d[A]}{dt} = k_1[A_3] - k_{-1}[A]^3$$

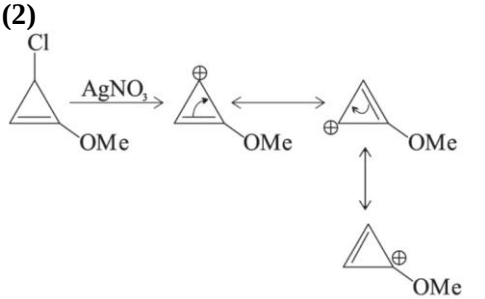
**29. (2)**




**30. (4)**

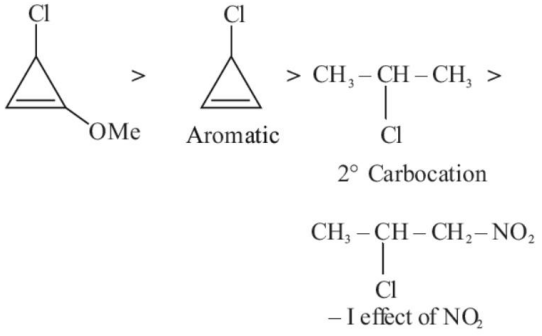
Cellulose is a homopolymer of β-D-glucose where $C_1$–$C_4$ linkage are present.

**31. (2)**




Aromatic as well as stabilized by lone pair of – OMe

So, the order is



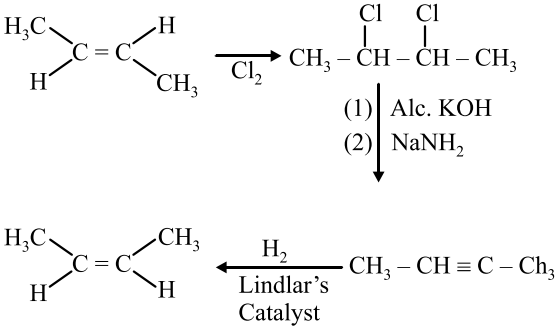

**32. (2)**

Inert pair effect gradually increases down the group.

Hence stability of lower O.S. increases and higher O.S decreases down the group.

**33. (2)**

The reaction takes place as:




Lindlar's catalyst makes cis-alkene as the major product.

**34.** **(1)**

$2\pi r = n\lambda$

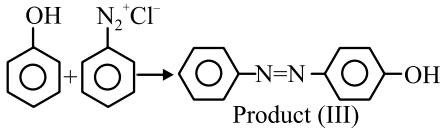
$$\lambda = \frac{2\pi r}{n} = \frac{2\pi n^2 a_0}{n \times Z} = 2\pi \frac{n}{Z} a_0$$

$\lambda = 2.5\pi a_0$

$$\therefore 2\pi \frac{n}{Z} a_0 = 2.5\pi a_0$$

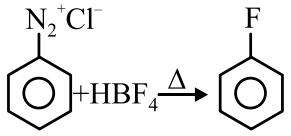
$$\therefore \frac{n}{Z} = \frac{2.5}{2}, \frac{z}{n} = \frac{2}{2.5} = 0.8$$

**35.** **(2)**

(A) 

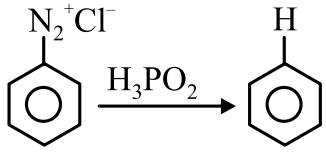
Product (III)

(Coupling reaction)

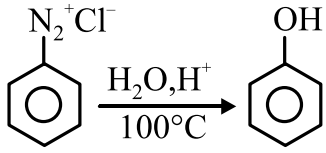
(B) 

Product(I)

(Balz-Schiemann reaction)

**(C)** 

Product(IV)

**(D)** 

Product(III)

**36.** **(4)**

$K_P = P_{CO_2}$ which will depend on temperature of the system only.

**37.** **(3)**

$\text{Rate} \alpha \frac{1}{\text{Steric Crowding}}$ keep this in denomentaor

also

**38.** **(2)**

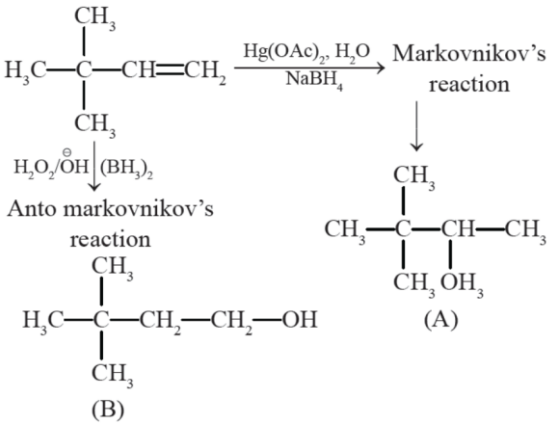
$$C_8H_{18}(l) + \frac{25}{2}O_2(g) \rightarrow 8CO_2(g) + 9H_2O(l)$$

$$\Delta n_g = n_p - n_r = 8 - \frac{25}{2} = -\frac{9}{2}$$

$\because \Delta H = \Delta U + \Delta n_g RT$

$$\therefore \Delta H - \Delta U = -\frac{9}{2}RT$$

**39.** **(4)**

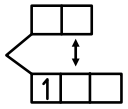



**40.** **(3)**

Since $Mg^{2+}$ is a small ion and carry more charge than $Mg^+$, therefore, more energy is required in its formation.

**41.** **(3)**

$[Ti(H_2O_6)]^{3+}$ $\quad Ti^{3+} \rightarrow d^1$




For octahedral:

CFSE = $-0.4\ \Delta_0$

**42.** **(1)**

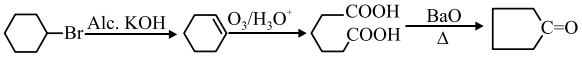
$NO_2$ group increases acidity due to its –M effect and $N(CH_3)_2$ decreases acidity due to its +M effect.

Hyperconjugation decreases acidity of isopropyl.

$\therefore$ the correct order of acidic strength is

(iii) > (i) > (iv) > (ii)

**43.** **(3)**




**44.** **(4)**

Given: ($K_f$ = 3K kg $mol^{-1}$, i= 3, Molarity = 0.05 mol $Kg^{-1}$)

Depression in freezing point is given as:

$\Delta T_f = i\ K_f m$

= 3 (3) (0.05)

**45.** **(3)**

| | $2SO_2(g) +$ | $O_2(g) \rightleftharpoons$ | $2SO_3(g)$ |
|---|---|---|---|
| Initial mol. | 2 | 1 | 0 |
| Eq. conc. | $\frac{(2-2x)}{2}$ | $\frac{(1-x)}{2}$ | $\frac{2x}{2}$ |

$K_2Cr_2O_7$ oxides, $SO_2$ remaining of equilibrium, Hence

Equivalent of $SO_2$ left = equivalents of $K_2Cr_2O_7$ used $(2 - 2x) \times 2 = (0.2 \times 0.1 \times 6)$

This gives, $x \approx 0.97$

Now, $K = \dfrac{\left(\frac{2x}{2}\right)^2}{\left(\frac{2-2x}{2}\right)^2 \times \left(\frac{1-x}{2}\right)} \approx 6.96 \times 10^4$

**46. (25)**

$$n_{C_3H_8} = \frac{200}{44}$$

$$N_{O_2} = \frac{2000}{32}$$

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$$

| | | | | |
|---|---|---|---|---|
| At t = 0; | 4.54 mol | 62.5 mol | 0 | 0 |
| At t = ∞; | 0 | 39.8 mol | 13.62 mol | 18.16 mol |

$$X_{H_2O} = \frac{18.16}{39.8 + 13.62 + 18.16} = 25.3 \times 10^{-3}$$

**47. (132)**

$$\text{Conductance(C)} = \frac{k}{x(\text{cell constant})}$$

$k = 0.66 \times 10^{-3} \times 1.3$

$$\Lambda_m = \frac{1000 \times 0.66 \times 10^{-3} \times 1.2}{6 \times 10^{-3}} = 132 \text{ Scm}^2\text{mol}^{-1}$$

**48. (0)**

The electronic configuration of Zn and $Zn^+$ are:

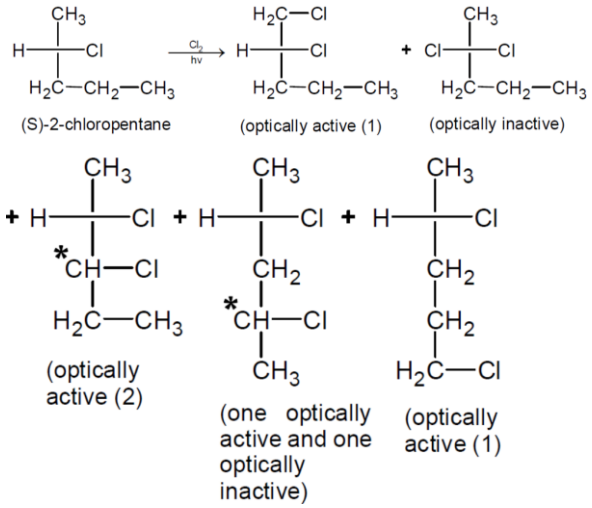
$Zn(30) = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}$,

$Zn^+ = [Ar]4s^1\ 3d^{10}$

Hence, the outermost electron is 4s electron, for which the following data is true.

$n = 4, l = 0, m = 0, s = \pm ½$

**49. (5)**




**50. (241)**

| | NaOH | + HCl | → | NaCl | + $H_2O$ |
|---|---|---|---|---|---|
| (Millimole) t = 0 | 400×0.5 | 600×1.0 | | 0 | 0 |
| t = ∞ | – | 400 | | 200 | 200 |

$$\text{Unreacted HCl mole} = \frac{400}{1000}$$

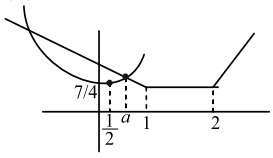
Number of unreacted HCl molecule

$$= \frac{400}{1000} \times 6.023 \times 10^{23}$$

$$= 240.92 \times 10^{21} \approx 241 \times 10^{21}$$

## SECTION-III (MATHEMATICS)

**51. (1)**




2 intersection points hence two solutions

**52. (1)**

A matrix A is symmetric if $A^T = A$ and skew-symmetric if $A^T = -A$ and its all diagonal elements are zeros and zero matrix is both symmetric and skew- symmetric

**53. (2)**

for sum of coefficient put $x = 1$

**54. (3)**

$$\arg(z-1) - \arg(z+1) = \frac{\pi}{4}$$

$$\arg(x+iy-1) - \arg(x+iy+1) = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{y}{x-1}\right) - \tan^{-1}\left(\frac{y}{x+1}\right) = \frac{\pi}{4}$$

$$\tan^{-1}\frac{\left(\frac{y}{x-1} - \frac{y}{x+1}\right)}{\left(1 + \frac{y}{x-1} \times \frac{y}{x+1}\right)} = \frac{\pi}{4}$$

$$\frac{yx + y - yx + y}{x^2 - 1 + y^2} = 1$$

$$2y = x^2 + y^2 - 1$$

**55. (3)**

$2x + y = 0$

and $x - y = 3$

Add $3x = 3$

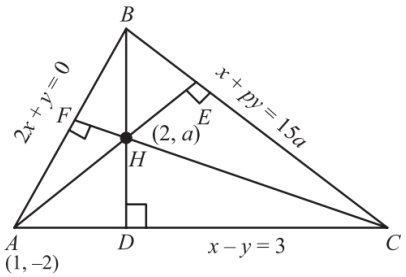
$\Rightarrow x = 1$

So, $y = -2$

For co-ordinate $C$

$3 + y + py = 15a \Rightarrow y = \dfrac{15a - 3}{p + 1}$

So, $x = \dfrac{15a + 3p}{p + 1}$




Slope of $AH = \dfrac{a + 2}{1}$

Slope of $BC = -\dfrac{1}{p}$

$\therefore p = a + 2 \quad \left[m_1 m_2 = -1\right] \qquad ...(i)$

**56. (3)**

$$\frac{\left(\hat{a}_2 - \hat{a}_1\right) \cdot \left(\hat{b}_1 \times \hat{b}_2\right)}{\left|\vec{b}_1 \times \vec{b}_2\right|}$$

$$\frac{x + \sqrt{6}}{2} = \frac{y - \sqrt{6}}{3} = \frac{z - \sqrt{6}}{4}$$

$$\frac{x + 2\sqrt{6}}{3} = \frac{y - 2\sqrt{6}}{4} = \frac{2 + 2\sqrt{6}}{5} \text{ is } 6$$

**57. (4)**

$\Rightarrow \quad 1 + 2 + 4 + 5 + x + y = 6 \times 5$

$\Rightarrow \quad x + y = 18 \ \{\because \text{Mean} = 5\} \qquad ....(i)$

$$\text{Variance } = 10 = \frac{1 + 4 + 16 + 25 + x^2 + y^2}{6} - (5)^2$$

$\Rightarrow \quad x^2 + y^2 = 164$

By solving (i) and (ii), we get

$x = 8, y = 10$

**58. (2)**

$n(s) = 36$

Given : $N - 2, \sqrt{3N}, N + 2$ are in G.P.

$3N = (N - 2)(N + 2)$

$3N = N^2 - 4$

$\Rightarrow N^2 - 3N - 4 = 0$

$(N - 4)(N + 1) = 0 \Rightarrow N = 4$ or $N = -1$ (rejected)

(Sum = 4) = {(1, 3), (3, 1), (2, 2)}

So, $n(A) = 3$

Hence,

$$P(A) = \frac{3}{36}$$

**59. (4)**

$y^2 = -4(x - a^2)$

Vertices of triangle are $(a^2, 0)$ and $(0, 2a)$ and $(0, -2a)$

$$\text{Area} = \frac{1}{2}\left(a^2\right)(4a) = \frac{1}{2}.4.125 = 250$$

**60. (1)**

$$f'(x) = \begin{cases} \cos x & , 0 < x < \dfrac{\pi}{2} \\ 0, & \dfrac{\pi}{2} < x < \pi \\ -\sin x, & x > \pi \end{cases}$$

$$f'\left(\frac{\pi^+}{2}\right) = 0$$

$$f'\left(\frac{\pi^-}{2}\right) = 0$$

$$f'\left(\pi^+\right) = 0$$

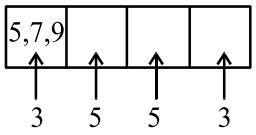
$$f'\left(\pi^-\right) = 0$$

$\Rightarrow f(x)$ is differentiable in $(0, \infty)$

**61. (4)**

Numbers between 5000 and 10000

Using digits 1, 3, 5, 5, 7, 9




$= 3 \times 5^3 = 375$

**62. (1)**

$$\tan^{-1}(x + 1) + \cot^{-1}\left(\frac{1}{x - 1}\right) = \tan^{-1}\frac{8}{31}$$

Taking tangent both sides

$$\frac{(x + 1) + (x - 1)}{1 - \left(x^2 - 1\right)} = \frac{8}{31}$$

$$\Rightarrow \frac{2x}{2 - x^2} = \frac{8}{31}$$

$$\Rightarrow 4x^2 + 31x - 8 = 0$$

$$\Rightarrow x = -8, \frac{1}{4}$$

But, if $x = \dfrac{1}{4}$

$\tan^{-1}(x+1) \in \left(0, \frac{\pi}{2}\right)$

$\cot^{-1}\left(\frac{1}{x-1}\right) \in \left(\frac{\pi}{2}, \pi\right)$

$\Rightarrow LHS > \frac{\pi}{2}$ and $RHS < \frac{\pi}{2}$

**63. (2)**

Both $R_1$ & $R_2$ are not transitive

**64. (1)**

Put $y = 0$ $\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$,

So, ellipse intersect x-axis at 7, i.e. a = 7

**65. (3)**

$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$

$\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$

$(\vec{r} - \vec{c}) \times \vec{b} = \vec{0}$

$\Rightarrow (\vec{r} - \vec{c})$ and $\vec{b}$ are parallel vectors

$\Rightarrow \vec{r} - \vec{c} = \mu\vec{b} \Rightarrow \vec{r} = \vec{c} + \mu\vec{b}$

Also, Also, $(\vec{c} + \mu\vec{b}) \cdot \vec{a} = 0$ ...(i)

$[\because \vec{r} \cdot \vec{a} = 0]$

$\Rightarrow \vec{a} \cdot \vec{c} + \mu(\vec{a} \cdot \vec{b}) = 0$

$\therefore \mu = \frac{-\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} = \frac{-8}{5}$

Now, from equ (i)

$\vec{r} = \frac{5(5\hat{i} - 3\hat{i} + 3\hat{k}) - 8(\hat{i} - \hat{j} + 2\hat{k})}{5}$

$\Rightarrow \vec{r} = \frac{17\hat{i} - 7\hat{j} - \hat{k}}{5}$

**66. (3)**

$3x^2 - 4y^2 = 36 \Rightarrow \frac{x^2}{12} - \frac{y^2}{9} = 1$

$\Rightarrow e = \sqrt{\frac{1+9}{12}} = \sqrt{\frac{21}{12}}$

$\frac{1}{\left(e'\right)^2} = 1 - \frac{1}{e^2} = 1 - \frac{12}{21}$

$\frac{1}{\left(e'\right)^2} = \frac{3}{7}$

$\Rightarrow e' = \sqrt{\frac{7}{3}}$

**67. (3)**

$\left(xy^3 - x^2\right)dy - \left(xy + y^4\right)dx = 0$

$\Rightarrow y^3\left(x\,dy - y\,dx\right) - x\left(x\,dy + y\,dx\right) = 0$

$\Rightarrow x^2y^3\frac{\left(x\,dy - y\,dx\right)}{x^2} - x\left(x\,dy + y\,dx\right) = 0$

$\Rightarrow x^2y^3\,d\left(\frac{y}{x}\right) - x\,d\left(xy\right) = 0$

$\Rightarrow$ Dividing by $x^3y^2$, we get

$\Rightarrow \frac{y}{x}d\left(\frac{y}{x}\right) - \frac{d\left(xy\right)}{x^2y^2} = 0$

Now integrating $\frac{1}{2}\left(\frac{y}{x}\right)^2 + \frac{1}{xy} = c$

It passes through the point $(4,-2)$

$\Rightarrow \frac{1}{8} - \frac{1}{8} = c \Rightarrow c = 0$

$\therefore y^3 = -2x$

**68. (1)**

$A = \int_0^{\frac{\pi}{4}}((\sin x + \cos x) - (\cos x - \sin x))dx$

$A = 2\int_0^{\frac{\pi}{4}}\sin x dx$

$A = -2\left(\frac{1}{\sqrt{2}} - 1\right)$

**69. (4)**

Given, $f(x) = ax^2 - 6x + 15$

Now, $f(x) = 2ax - 6 = 0$

For, maxima and minima $f'(x) = 0$

$\therefore 2ax - 6 = 0 \Rightarrow x = \frac{3}{a} = \frac{-3}{4} \Rightarrow a = -4$

$\therefore g(x) = -4x^2 + 6x - 15$

**70. (3)**

$\left(3^{1/2} + 5^{1/8}\right)^{256}$

$T_{r+1} = {}^nC_r(3)^{\frac{256-r}{2}}(5)^{\frac{r}{8}}$

$\therefore r$ shall be multiple of 8

**71. (1)**

Given, $\frac{10}{\pi}\int_0^{\sqrt{2}}\frac{\left(2 - x^2\right)}{\left(x^2 + 2\right)\sqrt{4 + x^4}}dx$

$$=\frac{10}{\pi}\int_0^{\sqrt{2}}\frac{x^2\left(\frac{2}{x^2}-1\right)dx}{x\left(x+\frac{2}{x}\right)\times x\sqrt{\frac{4}{x^2}+x^2}}$$

$$=\frac{10}{\pi}\int_0^{\sqrt{2}}\frac{\left(\frac{2}{x^2}-1\right)dx}{\left(x+\frac{2}{x}\right)\sqrt{\left(x+\frac{2}{x}\right)^2-4}}$$

put $x+\frac{2}{x}=t$

$\Rightarrow dt=\left(1-\frac{2}{x^2}\right)dx$

$\Rightarrow I=-\frac{10}{\pi}\int\frac{dt}{t\sqrt{t^2-4}}$

$$=-\frac{10}{\pi}\times\frac{1}{2}\sec^{-1}\frac{x+\frac{2}{x}}{2}\Bigg|_0^{\sqrt{2}}$$

$$=-\frac{5}{\pi}\left[\sec^{-1}\left(\frac{2\sqrt{2}}{2}\right)-\sec^{-1}(\infty)\right]=\frac{5}{4}$$

**72. (2)**

$x^2+y^2+(x-1)^2+y^2+x^2+(y-1)^2+(x-1)^2+$

$(y-1)^2=18$

$\Rightarrow x^2+y^2-x-y-\frac{7}{2}=0$

**73. (6)**

$I=\int\frac{\sin x}{\sin^3 x+\cos^3 x}dx$

$I=\int\frac{\tan x\sec^2 x}{\tan^3 x+1}dx$

Let, $\tan x = t$

$\text{Sec}^2\, x\, dx = dt$

$\int\frac{tdt}{t^3+1}=\int\frac{t}{(t+1)\left(t^2-t+1\right)}dt$

$\frac{t}{(t+1)\left(t^2-t+1\right)}=\frac{A}{(t+1)}+\frac{Bt+C}{t^2-t+1}$

$t=A\left(t^2-t+1\right)+(Bt+C)(t+1)$

$A=\frac{-1}{3},B=\frac{1}{3},C=\frac{1}{3}$

$I=\frac{-1}{3}\int\frac{1}{t+1}dt+\frac{1}{3}\int\frac{t+1}{t^2-t+1}d\ dt$

$=\frac{-1}{3}\ln(\tan x+1)+\frac{1}{6}\ln\left(\tan^2 x-\tan x+1\right)$

$+\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2\tan x-1}{\sqrt{3}}\right)+c$

**74. (39)**

$$=\frac{\left(\frac{{}^{12}C_2}{{}^{51}C_2}\right)\times\frac{1}{4}}{\frac{{}^{12}C_2}{{}^{51}C_2}\times\frac{1}{4}+\left(\frac{{}^{13}C_2}{{}^{51}C_2}\right)\times\frac{3}{4}}=\frac{11}{50}$$

**75. (64)**

$=\left(\frac{1}{4}\sin 30^\circ\right)\cdot\frac{1}{4}\cos 30^\circ$

$=\frac{\sqrt{3}}{64}$



**PW Web/App** - https://smart.link/7wwosivoicgd4

**Library-** https://smart.link/sdfez8ejd80if